

Aangan — Procurement List

Nathaniel School of Music, Bangalore · two orders, kept as two separate invoices · Silverline: WhatsApp +91 70909 39819 before visiting (139/5 VT Complex, opp. Karthik Plaza, Nagarathpete, SP Road) · prices are budget references, confirm with vendor

Already in hand — not part of this order: Raspberry Pi 5 8GB · ESP32 x6 · ESP32-CAM x1 · HC-SR04 ultrasonic x1 · buzzer modules x4 · MEMS/INMP441 mic x1 · soil-moisture sensor x1 · DHT22 x1 · water-leak sensor x1 · WS2812B LED strip x1 · jumper wires x6 · old iPad · certified smoke + LPG alarms · USB A→B cable · XLR cables · USB-C 100W cables · USB SSD · gaffer + velcro · studio UPS.

ORDER 1 — Studio & House system

A · SILVERLINE ELECTRONICS (SP ROAD)

ITEM	QTY	REF. PRICE	LINK
Raspberry Pi 5, 8GB	1	₹8,000	silverline — Pi 5 8GB
Official Raspberry Pi 27W USB-C power supply	1	₹1,300	silverline — 27W PSU

B · DOOR SCREENS (AMAZON.IN)

ITEM	QTY	REF. PRICE	LINK
Samsung Galaxy Tab A9, 8.7", Wi-Fi front-door display + studio-door display (kiosk mode)	2	₹10,000 ea	amazon — Tab A9
Screw-in tablet wall mount, fits 8.7" tablet	2	₹700 ea	amazon — wall mount
Flat USB-C cable 2m + 2A wall charger	2	₹500 ea	amazon — cable + charger
TP-Link Tapo C210 camera (3MP) entrance camera	1	₹2,800	amazon — Tapo C210

C · SENSORS & ELECTRONICS (SP ROAD COMPONENTS STORES; ROBU.IN LINKS FOR REFERENCE/FALLBACK)

ITEM	QTY	REF. PRICE	LINK
MC-38 wired reed door switch (NC, with magnet)	12	₹50 ea	MC-38
Water-leak sensor module	9	₹80 ea	leak sensor
DHT22 temperature/humidity module	4	₹290 ea	DHT22
MQ-6 LPG gas sensor module	4	₹120 ea	MQ-6
IR flame sensor module	4	₹70 ea	flame sensor
PIR motion sensor HC-SR501	4	₹70 ea	PIR

SW-420 vibration sensor module	5	₹70 ea	SW-420
HLK-LD2410 mmWave presence sensor	1	₹350	LD2410
DFRobot Gravity SEN0232 sound level meter	1	₹3,500	SEN0232
JSN-SR04T waterproof ultrasonic sensor	1	₹450	JSN-SR04T
HX711 amplifier + 50kg load cells, set of 4	1 set	₹400	HX711 set
Momentary push button, panel mount doorbell button	1	₹30	push button
Jumper wire ribbon pack, 120 pc (M-M / M-F / F-F)	1	₹350	jumpers
Solderless breadboard, 840 point	3	₹150 ea	breadboard
5V 2A USB wall charger + micro-USB cable one per ESP32 node	6	₹200 ea	SP Road / local
SanDisk Extreme A2 microSD 64GB	1	₹700	SP Road / local

D · ELECTRICAL (SP ROAD ELECTRICAL STORES)

ITEM	QTY	REF. PRICE	LINK
Latching panic switch, Schneider-type	4	₹350 ea	electrical store
2-core security loop cable, 90m roll	1	₹1,200	electrical store
Battery-backed alarm sounder, 12V	1	₹1,500	electrical store
Volt-free relay module (for certified-alarm contacts)	3	₹250 ea	electrical store
25A optoisolated relay module geyser switching	4	₹450 ea	25A relay
Motor contactor + float switches (water pump)	1 + 2	₹2,000	electrical store

ORDER 2 — Music rig uses the Raspberry Pi 5 already in hand · runs BOTH Pianoteq and Chorale

Board change — read before ordering. The Raspberry Pi DAC Pro and its XLR daughter board are **output-only**. Pianoteq is happy on them, but Chorale is a harmonizer — it needs an analogue *input*, and that board has none. Since both apps are meant to run on this Pi, the audio board must have an ADC. Only one audio HAT can occupy the I2S bus, so this is a choice, not an addition.

A · AUDIO CORE — THE ADC DECISION

ITEM	QTY	REF. PRICE	LINK
HiFiBerry DAC2 ADC Pro RECOMMENDED. Stereo line in + line out, 24-bit/192k, input stage built for studio line level. One board serves Pianoteq (out) and Chorale (in + out). Still a HAT — nothing external, nothing USB.	1	₹7,500 + shipping	hifiberry.com

Behringer DI400P passive DI The ADC Pro's output is RCA; these restore balanced XLR into the console.	2	₹1,000 ea	amazon — DI400P
Raspberry Pi DAC Pro + XLR board ONLY if the Pi stays a Pianoteq-only rig and Chorale runs on the Mac. Do NOT buy alongside the HiFiBerry — they cannot coexist on the same Pi.	(1)	₹4,500	robu — DAC Pro
6-pin female header, 2.54mm	2	₹30	header

Why there is no mic preamp on this list: on stage the console already has the preamp and the phantom power. Take a line-level send from the vocal channel into the Pi, return the harmonies to two spare channels. Line in, line out — exactly what this board is designed for.

A2 · STAGE-WORTHINESS — THE PARTS THAT STOP A PI FAILING IN FRONT OF AN AUDIENCE

ITEM	QTY	REF. PRICE	LINK
True-bypass looper pedal, 1 loop NON-NEGOTIABLE. A mechanical relay that lifts the Pi out of the signal path with one stomp, so a crash or a reboot is a five-second recovery instead of silence in front of an audience.	1	₹2,500	amazon — true bypass looper
NVMe HAT + 256GB NVMe SSD Boot from SSD, not microSD. Card corruption on power loss is the most common way a Pi dies in the field, and the card is the part that dies.	1	₹4,500	robu — NVMe HAT
UPS HAT with 18650 cells Rides out a brownout or a pulled cable long enough to shut down cleanly.	1	₹3,500	robu — UPS HAT
Raspberry Pi Pico (RP2040) The footswitch brain — no OS, no card, USB-MIDI in under a second. Firmware is written: chorale repo, tools/pico-footswitch/code.py. Buy two so there is a spare in the case.	2	₹400 ea	robu — Pico
Momentary SPST footswitches Harmony · Voices · Mode · Doubler · Stage/Studio · Panic	6	₹120 ea	robu — footswitch
1590BB aluminium enclosure + 6.35mm jacks Metal, not plastic — it gets stood on.	1	₹1,200	robu — enclosure
Expression pedal + TRS cable Optional, but the most musical control on the board — swells the choir on CC 11.	(1)	₹3,500	amazon — expression pedal

B · CONTROL, COOLING, POWER, STORAGE

ITEM	QTY	REF. PRICE	LINK
Raspberry Pi Touch Display 2, 7"	1	₹5,500	silverline — Touch Display 2
Official Raspberry Pi 5 Active Cooler	1	₹500	silverline — Active Cooler
Official Raspberry Pi 27W USB-C power supply	1	₹1,300	silverline — 27W PSU
SanDisk Extreme A2 microSD 64GB one runs the rig, one is the cloned spare	2	₹700 ea	SP Road / local

100W USB-C PD power bank, 20,000mAh+	1	₹10,000	amazon — PD bank
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△ **Set Chorale's Trigger CC to 27 before the first gig.** It defaults to CC 64, which is also the MIDI sustain-pedal message. If a keyboard shares the MIDI port, its sustain pedal will silently toggle your harmony on and off mid-song. Change it in Chorale and set TRIGGER_CC = 27 in the pedal firmware to match.

C · MIDI

ITEM	QTY	REF. PRICE	LINK
Roland UM-ONE mk2 USB-MIDI interface	1	₹4,000	amazon — UM-ONE mk2
CME WIDI Master wireless MIDI	1	₹9,000	amazon — WIDI Master

D · TRANSPORT

ITEM	QTY	REF. PRICE	LINK
Hard foam-lined equipment case size finalised after the rig is assembled	1	₹3,000	music store

ORDER 3 — Air, ventilation & environment the existing Dyson integrates at no cost; this is the layer around it

A · PURIFIERS (MI.COM/IN)

ITEM	QTY	REF. PRICE	LINK
Xiaomi Smart Air Purifier 4 bedroom	1	₹13,000	mi.com — Purifier 4
Xiaomi Smart Air Purifier 4 Lite teaching room	1	₹10,000	mi.com — Purifier 4 Lite

B · AIR-QUALITY SENSORS (SP ROAD COMPONENTS STORES; ROBU.IN LINKS FOR REFERENCE)

ITEM	QTY	REF. PRICE	LINK
PMS5003 laser particulate sensor (PM2.5/PM10)	3	₹2,000 ea	PMS5003
SCD41 CO ₂ sensor module (true NDIR/photoacoustic)	3	₹3,500 ea	SCD41
SGP41 VOC + NOx index sensor module	3	₹2,000 ea	SGP41
SHT45 temperature/humidity module	2	₹1,000 ea	SHT45
TDS water quality probe RO drinking line	1	₹600	TDS sensor
ABS project box, 100×68×50mm one per air node	3	₹200 ea	SP Road

C · ELECTRICAL (SP ROAD ELECTRICAL STORES)

ITEM	QTY	REF. PRICE	LINK
Relay module for exhaust fan switching kitchen + 2 bathrooms	3	₹250 ea	electrical store

Aangan · Nathaniel School of Music · rev 5 (31 Jul 2026) · Order 1 ≈ ₹51,000 · Order 2 ≈ ₹31,500 core (stage-worthy, both apps) · Order 3 ≈ ₹47,000 (stageable — Dyson integration is ₹0; first proving step is one purifier + one air node ≈ ₹20,500) · build documents: “Aangan — Install Guide + Idea Menu”, “Aangan — Air, Ventilation & Environment Plan”, PIANO_PI_BUILD_PLAN.html in the aangan repo.